

1(a)(i)	<i>independent variable:</i> type of leaf / type of plant / type of disc / species (of plant) ; <i>dependent variable:</i> time for (leaf) discs to, rise to / reach, the surface / the top ;	2	I rate of photosynthesis
1(a)(ii)	<i>any 7 of:</i> 1 <i>ref. to</i> a method of transferring leaf discs ; 2 same, height / volume / depth, of solution / NaHCO₃, in beaker (for all leaf types) ; 3 use, timer / stop clock, to find time for leaf disc(s) to reach, surface / top ; 4 <i>ref. to</i> using same number of leaf discs (for all leaf types) ; 5 <i>ref. to</i> the same concentration of solution (for all leaf types) ; 6 <i>ref. to</i> standardising some aspect of the discs ;	7	<i>max. 6 if critical step mp9 missing</i> 1 e.g. forceps, glass rod, paint brush, tweezers 2 A other suitable containers e.g. syringe / boiling tube / test tube 3 A time for first leaf disc / all leaf discs / stated number of leaf discs / each one in a separate beaker R if different types of leaf in same beaker at same time 4 minimum of 3 6 e.g. same, size / diameter / age / position in leaf sampled

Question	Answer	Marks	Guidance
1(a)(ii) cont	7 <i>ref. to</i> keeping same light intensity ; 8 <i>ref. to method</i> for keeping temperature of solution constant ; <i>critical step</i> 9 start timer (immediately) when disc(s) reach bottom (of beaker) ; 10 low / medium, risk, experiment / investigation / procedure / assessment / AW ; 11 <i>ref. to</i> minimum of 3 readings (for at least one leaf type) and calculating a mean or excluding / identifying, anomalies ;		7 A methods that achieve same light intensity, e.g. lamp(s) at fixed distance / same wattage (bulbs) / variable resistor / dimmer 8 e.g. temperature-controlled room or chamber / environmental chamber / incubator / water-bath / thermostatic control I air conditioning / heat shield 10 R no risk I 'low hazard' experiment. A Allergy to leaves and wear, gloves / protective clothing A NaHCO₃ is a mild skin irritant to and wear, gloves / protective clothing I <i>ref. to</i> water and electricity / care with cutting discs 11 A having at least 3 discs in one beaker A one disc in 3 separate beakers I average / average mean A mean average
1(b)(i)	divide 1 by the time ;	1	A distance disc travels divided by time A divide 10 / 100 / 1000 by time I volume of oxygen divided by time

Question	Answer	Marks	Guidance
1(b)(ii)	<i>idea that:</i> 1 leaves (with thick cuticles) may be heavier which will, increase time to rise / require more oxygen to raise them or cuticles, are transparent / do not affect light absorption or thick cuticle only on upper surface so light absorption via lower surface not affected or difference between spiderwort with thin cuticle and sorghum with thick cuticle, is small / may not be significant or leaves (with thick cuticles may) have fewer stomata so less gas exchange ; 2 the purple parts of the leaf may contain chlorophyll (which is hidden by the purple pigment) / cannot assume that the purple stripes do not contain chlorophyll or no actual data on the chlorophyll content or the purple pigments may also absorb light (<i>idea of accessory pigments</i>) or spiderwort rate, close to sorghum / faster than ivy, which does not have purple stripes ; 3 experiment carried out at, 20 °C / room / low temperature, and C4 plants / Sorghum, only efficient at high(er) temperatures / ora ;	3	1 A in terms of leaves being thicker A no evidence that thick cuticles reduce light absorption I ref. to no lower cuticle A with ref. to carbon dioxide in / oxygen out 2 A idea that green stripes are more chlorophyll dense (than other leaves) could relate to purple stripes or whole leaf

Question	Answer	Marks	Guidance
1(c)(i)	1 <i>idea of</i> using same, quantity / amount, of leaf tissue from each plant ; e.g. (weigh to) obtain same mass / same number of discs / cut same area of leaf 2 <i>ref. to</i> outline method of extracting pigments ; e.g. (crush / leave, the leaves in) solvent / acetone / propanone / ethanol / alcohol or extraction solvent 3 <i>ref. to</i> using red filter (in a colorimeter) ; 4 <i>idea that</i> taking a reading for, absorbance / transmission, to compare each extract ;	3	1 A dry mass 2 I water as a solvent 3 A <i>idea of</i> shining a red light through the extract I 'chlorophyll absorbs red light' A if leaf / discs are used e.g. 'spiderwort chlorophyll has a lower / the lowest absorbance (of red light in the colorimeter)'
1(c)(ii)	solvent / named solvent (used to extract the chlorophyll) ;	1	A water

Question	Answer	Marks	Guidance
1(d)	<i>max 3 if only mp1-4 given</i> <i>use of t-test:</i> 1 <u>3</u> ; 2 <i>idea of</i> carrying out (separate) <i>t</i>-test on (pairs of data from) spiderwort and (each of) the other (3) plants / AW ; <i>finding the significance:</i> 3 calculate / find / use, the degrees of freedom / <i>v</i> ; 4 <i>ref. to</i> critical / table, value at, 0.05 (probability/significance) or 5% (confidence, level / index. limit) ; 5 if (calculated) value of <i>t</i>, is greater than / > , critical / table, value, the difference is significant ; ora	4	2 I value stated in mp1 3 must be in correct context A calculation (10–1)+(10–1) A 20 – 2 or statement that <i>v</i> is 18 I formula, as given 4 <i>ref. to</i> using 0.05 / 5% (column) R 0.05% A at 95% / 0.95 confidence limit / index 5 A in terms of results (of <i>t</i>-tests) were significant I ‘values’ are significant